# Model Documentation

# Risk Calculation System

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# 1 Executive Summary

This document describes the model used in the course risk calculation system and the design decisions reflected in the Python implementation. The system accepts a portfolio of stock and option positions, calibrates statistical inputs from historical data or from user-supplied parameters, computes value at risk (VaR) by historical simulation, parametric approximation, and Monte Carlo simulation, computes expected shortfall (ES) by historical and Monte Carlo methods, and backtests one-day VaR forecasts against realized portfolio outcomes.

The software is implemented as a class-based risk engine in Python. The central class, RiskCalculationSystem, is parameterized by the portfolio, the historical market data set, the confidence level, and the trading-day convention. Risk reports are produced through method calls that allow the user to select the lookback window, horizon, simulation count, and historical shock convention. This design keeps the software aligned with the project requirement that the system both calibrate to historical data and accept parameters as input.

The model is intentionally modular rather than overly ambitious. Historical and Monte Carlo methods use full portfolio revaluation, which is important for portfolios containing options. Parametric VaR and ES use a delta-normal approximation, which makes the calculation fast and transparent but introduces approximation error for strongly nonlinear portfolios. Option positions are priced with the Black-Scholes formula for European options. This gives a coherent and testable baseline implementation, even though it does not cover early exercise, volatility smile dynamics, or more complex products.

- Key strengths: transparent architecture, direct mapping from lecture formulas to code, support for both historical calibration and user-supplied parameters, and full revaluation for nonlinear positions in historical and Monte Carlo methods.

- Main limitations: options are restricted to European calls and puts; the parametric method is only first-order accurate for nonlinear portfolios; and the backtest currently supports one-day VaR only.

# 2 Introduction

The project requirement is to build a coherent risk calculation system rather than a collection of isolated homework scripts. Accordingly, the implementation described here standardizes the portfolio input format, centralizes valuation logic, separates calibration from risk measurement, and exposes each risk method through a consistent interface. The present document focuses on model documentation.

The functions reviewed in this documentation are implemented in the Python file risk_system.py. The intended use of the system is educational and prototype-oriented: it is designed to show how a small but disciplined market-risk engine can be built for a portfolio of stocks and European options. It is not presented as a production trading system. The design therefore prioritizes readability, modularity, and correctness of core calculations over optimization, market data infrastructure, and support for a broad product set.

The documentation follows the structure of the model validation template supplied by the professor. Where the template refers to validation, this document interprets validation in a practical course-project sense: functional verification, methodological review, and preliminary reasonableness checks rather than a full independent model risk review.

# 3 Product Description

The product covered by the system is a portfolio consisting of positions in common stocks and European options written on those stocks. Each portfolio row contains an instrument type, an identifier, a signed quantity, and, for options, the underlying symbol, call or put flag, strike, maturity, volatility input if supplied, and risk-free rate.

For stock positions, the payoff is linear in the underlying spot price. For option positions, the payoff is nonlinear and depends on the terminal underlying price relative to the strike. The software values options using the Black-Scholes model for European exercise. This choice is appropriate for a first implementation because it provides closed-form prices and deltas and allows the risk engine to support both full revaluation and a parametric approximation.

The historical market data input is a date-indexed price matrix for the underlying risk factors. The code assumes strictly positive prices so that log-return calibration is well defined. A separate portfolio template is supplied to standardize user input.

# 4 Model Description

## 4.1 Modeling theory and assumptions

The lecture notes define the p-level value at risk as the p-quantile of the loss distribution. Throughout the system, portfolio loss is measured by L = V0 - VT, so losses are positive when the future portfolio value falls below the current portfolio value. The software follows the same loss-based convention used in class and treats VaR as a quantile of the empirical or model-implied loss distribution.

$$\mathrm{VaR}(V, T, p) = G^{-1}(p), \qquad G(x) = P(V_0 - V_T \leq x) \tag{4.1}$$

$$\mathrm{ES}(V, T, p) = E[V_0 - V_T \mid V_0 - V_T > \mathrm{VaR}(V, T, p)] \tag{4.2}$$

Three VaR methods are implemented. Historical VaR applies observed historical shocks to the current market state and fully revalues the portfolio under each scenario. Parametric VaR assumes an approximately normal one-period P&L distribution and computes risk from the mean and variance of a delta-approximated portfolio. Monte Carlo VaR simulates correlated log returns under a multivariate normal model calibrated from historical data and then fully revalues the portfolio.

The main modeling assumptions are as follows. First, the underlying risk factors are equity spot prices only. Second, historical calibration uses either simple returns or log returns derived from the historical price panel. Third, option repricing uses Black-Scholes with constant volatility over the horizon unless a volatility is omitted, in which case annualized historical volatility is used as a fallback estimate. Fourth, for the Monte Carlo method, log returns are assumed jointly normal over the chosen horizon. Fifth, the backtesting framework compares one-day realized losses with one-day VaR forecasts from a rolling historical window.

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t, \qquad S_T = S_0 \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)T + \sigma W_T\right] \tag{4.3}$$

The Black-Scholes option pricer used in the software is based on the standard European call formula and its associated delta.

$$C = S_0 N(d_1) - K e^{-rT} N(d_2), \qquad d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T} \tag{4.10}$$

The most important approximation is in the parametric method. The code converts the portfolio to first-order spot exposures using stock positions and option deltas, then applies a delta-normal approximation to one-day P&L. This is computationally efficient, but it ignores gamma and vega effects and may materially understate or overstate tail risk for portfolios with large nonlinear positions.

## 4.2 Mathematical description

Historical simulation applies observed historical market moves to the current market state and then fully revalues the portfolio under each scenario. This is the most direct implementation of the class definition because no parametric shape is imposed on the loss distribution.

$$S^{(i)} = S_0 \odot e^{r^{(i)}} \text{ or } S^{(i)} = S_0 + \Delta S^{(i)}, \qquad L_i = V_0 - V^{(i)} \tag{4.4}$$

$$\mathrm{VaR}_p^{\mathrm{hist}} = \mathrm{Quantile}_p(\{L_i\}_{i=1}^n), \qquad \mathrm{ES}_p^{\mathrm{hist}} = E[L_i \mid L_i > \mathrm{VaR}_p^{\mathrm{hist}}] \tag{4.5}$$

Parametric VaR and ES are implemented through a delta-normal approximation. The portfolio is linearized with respect to current spot levels, so the one-period profit and loss is approximated by a normal random variable determined by the first-order exposure vector and the sample moments of returns.

$$\Delta P \approx (\Delta \odot S_0)^\top r, \qquad \mu_P = (\Delta \odot S_0)^\top \mu, \qquad \sigma_P^2 = (\Delta \odot S_0)^\top \Sigma (\Delta \odot S_0) \tag{4.6}$$

$$\mathrm{VaR}_p^{\mathrm{par}} = -h\mu_P + z_p\sqrt{h}\,\sigma_P, \qquad \mathrm{ES}_p^{\mathrm{par}} = -h\mu_P + \sqrt{h}\,\sigma_P \frac{\phi(z_p)}{1-p} \tag{4.7}$$

Monte Carlo VaR and ES simulate correlated return scenarios from a multivariate normal model calibrated to historical log returns. Unlike the parametric approximation, each scenario is passed through the full pricing logic, so nonlinear option payoffs are preserved.

$$R^{(m)} \sim N(h\mu, h\Sigma), \qquad S^{(m)} = S_0 \odot e^{R^{(m)}}, \qquad L_m = V_0 - V^{(m)} \tag{4.8}$$

Backtesting is performed for one-day VaR forecasts by rolling a fixed estimation window through time. At each forecast origin, the chosen model is recalibrated using only information available up to that date, a one-day VaR is produced, and the realized next-day loss is compared with the forecast threshold. The summary reports the number of exceptions, the observed exception rate, and the Kupiec unconditional coverage statistic.

$$\mathrm{LR}_{uc} = -2\log\left[\frac{(1-p)^{N-x}p^x}{(1-\hat{p})^{N-x}\hat{p}^x}\right], \qquad \hat{p} = \frac{x}{N} \tag{4.9}$$

## 4.3 Model inputs and outputs

| Item | Description |
|---|---|
| Portfolio input | CSV or DataFrame containing instrument_type, symbol, quantity, and option fields when applicable. |
| Historical market data | Date-indexed price history for each risk symbol; must be positive and include all required underlyings. |
| Core user parameters | Confidence level, lookback window, horizon in trading days, number of Monte Carlo simulations, and historical shock type. |
| Optional model parameters | User-supplied mean vector, covariance matrix, spot levels, option volatilities, and rates. |
| Primary outputs | Current portfolio value, VaR, ES when applicable, number of scenarios or simulations, and backtest diagnostics. |

## 4.4 Model implementation and numerical methods

The software is organized around three core classes or structures: Position, BlackScholesPricer, and RiskCalculationSystem. Position stores one standardized portfolio row. BlackScholesPricer provides price and delta methods for European options. RiskCalculationSystem aggregates positions, historical prices, calibration routines, valuation logic, risk measures, and backtesting utilities.

The implementation deliberately favors an object-oriented interface. A single class instance stores the portfolio, price history, current spots, and derived return series. Risk calculations are then method calls on that instance, which makes the design easy to document and test. The public methods from_csv, calibrate_from_history, portfolio_value, historical_var_es, parametric_var_es, monte_carlo_var_es, and backtest_var correspond closely to the project requirements.

The major numerical issues are straightforward but important. Historical and Monte Carlo methods require repeated full revaluation of option positions across many scenarios; this is accurate for the supported product set but slower than linear methods. The Monte Carlo simulation count affects the stability of tail estimates, especially ES. The historical method depends materially on the lookback window and on whether shocks are applied relatively or absolutely. The parametric method is fast but only as reliable as the normal and first-order approximations permit.

## 4.5 Calibration methodology

Calibration is intentionally simple and transparent. The system estimates daily means, covariances, annualized volatilities, and correlations directly from historical return samples. The user may choose whether those samples are based on log returns or simple returns, depending on the downstream method. In addition, the parametric and Monte Carlo methods accept user-supplied mean vectors and covariance matrices so that the system can be run with externally calibrated parameters rather than only sample estimates.

For option volatility, the preferred approach is for the user to provide a volatility in the portfolio file. If no volatility is supplied, the code falls back to an annualized historical volatility estimate for the underlying. This fallback keeps the system operable, but it should be viewed as a convenience rather than a substitute for a proper implied-volatility input in a more realistic derivatives setting.

# 5 Validation Methodology and Scope

The scope of validation in this draft is limited to what can be supported directly by the implemented software and the project materials. The review therefore focuses on internal consistency, reasonableness of formulas, alignment between documented methodology and code, and basic smoke testing of the risk methods. This is not a full independent benchmark study against a commercial risk platform.

- Code and formula review: each method was checked against the lecture definitions of loss, VaR quantile, and expected shortfall.
- Input and output review: portfolio parsing, historical price validation, and scenario-based valuation were examined to ensure that required fields and assumptions are explicit.
- Functional testing: example portfolios and simulated price histories were used to verify that all major methods run end-to-end and return economically sensible outputs.
- Backtesting review: the rolling one-day framework and Kupiec statistic were reviewed to confirm that the exception logic matches the intended interpretation of VaR breaches.

A full validation performed later in the course should go further. In particular, it should compare historical, parametric, and Monte Carlo outputs under controlled test portfolios; verify convergence of Monte Carlo estimates as the number of simulations increases; stress edge cases such as deep in-the-money and near-expiry options; and compare backtest behavior across different confidence levels and windows.

# 6 Validation Results

Preliminary smoke tests were run on an example portfolio containing a long AAPL stock position, a long AAPL call option, and a short MSFT stock position. A synthetic but realistic price history with 400 business dates was generated from a correlated two-asset lognormal process. The purpose of this exercise was not to claim empirical accuracy, but to check that the model produces stable outputs and that the three VaR methods behave in the expected order of magnitude.

| Method | Reported quantity | Result |
|---|---|---|
| Current portfolio value | V0 | $26,250.50 |
| Historical simulation | 99% VaR / ES | $1,565.53 / $1,787.54 |
| Parametric delta-normal | 99% VaR / ES | $1,934.67 / $2,244.04 |
| Monte Carlo full revaluation | 99% VaR / ES | $1,900.55 / $2,186.94 |
| Historical backtest | Exceptions over 147 observations | 0 exceptions |
| Kupiec coverage test | LR_uc / p-value | 2.955 / 0.086 |

The outputs are directionally reasonable. The historical result is somewhat lower than the parametric and Monte Carlo values, which is plausible given the particular simulated sample. The parametric and Monte Carlo estimates are close enough to suggest that, for this test portfolio, the delta-normal approximation is not grossly inconsistent with the full-revaluation simulation. The backtest produced no exceptions over 147 one-day forecasts. This does not prove superiority; it merely indicates that the historical VaR forecast was not obviously too low in the sample tested.

These results should be interpreted as preliminary verification only. Because the test data are simulated and the portfolio is small, they do not establish production-grade accuracy. They do, however, support the claim that the implementation is internally coherent, runs end-to-end, and produces economically sensible risk measures.

# 7 Conclusions and Recommendations

The current model documentation supports a clean first version of the course risk calculation system. The design is appropriate for the project requirements because it accepts a portfolio of stocks and options, calibrates from historical data while also allowing direct parameter inputs, computes historical, parametric, and Monte Carlo VaR, computes historical and Monte Carlo ES, and backtests VaR against realized one-day outcomes.

- Recommendation 1: retain full revaluation for historical and Monte Carlo methods, since this is the most defensible design choice for portfolios with nonlinear payoffs.
- Recommendation 2: keep the parametric method clearly labeled as a delta-normal approximation and document that it is primarily a fast benchmark rather than a definitive risk number for option-heavy books.
- Recommendation 3: in a later version, add stressed VaR or stressed historical windows, richer option support, and explicit convergence diagnostics for Monte Carlo ES.
- Recommendation 4: extend the testing framework with deterministic unit tests, benchmark portfolios with known analytical behavior, and multi-asset backtests at several confidence levels.

Overall, the implementation is suitable as a course project deliverable and provides a strong foundation for the remaining documents, including the software design documentation, test plan, and test results.

# 8 Bibliography


- Course lecture slides on VaR, Expected Shortfall, GBM VaR, Parametric VaR, Monte Carlo VaR, and Historical VaR.
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- Jorion, P. (2007). Value at Risk: The New Benchmark for Managing Financial Risk. 3rd ed. McGraw-Hill.
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# Appendix A Software Mapping

The Python software file risk_system.py contains the following user-facing components: Position dataclass for standardized portfolio rows; BlackScholesPricer for European option pricing and delta; RiskCalculationSystem class for calibration, valuation, VaR, ES, and backtesting; and convenience helpers for a sample portfolio and example usage. This appendix is included so that the model documentation points directly to the software structure used in risk_system.

| Software component | Purpose |
|---|---|
| from_csv | Loads the standardized portfolio and price-history inputs from CSV files. |
| calibrate_from_history | Computes sample means, covariance, annualized volatility, and correlation from historical returns. |
| portfolio_value | Computes current portfolio value under supplied or stored spot prices. |
| historical_var_es | Computes historical simulation VaR and ES using relative or absolute historical shocks. |
| parametric_var_es | Computes delta-normal VaR and ES using first-order exposures and sample or supplied moments. |
| monte_carlo_var_es | Computes Monte Carlo VaR and ES by simulating correlated log returns and fully repricing the portfolio. |
| backtest_var | Runs a rolling one-day backtest and reports exception counts and Kupiec coverage statistics. |